# Kurella Pavan

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 7.11 |
| Narayana College | Class XII (Telangana State Board of Intermediate Education) | 2023 | 98.30% |
| Abhyas Techno High School | Class X (Telangana State Board of Secondary Education) | 2021 | 10.00% |

## PROJECTS

**Operating System with Scheduling, Virtual Memory & Storage Features**

*C++ | xv6 | Scheduling | Memory Management*

- **Extended the xv6 RISC-V kernel** with custom **system calls** for process relationships, system-call accounting, scheduler inspection, and virtual-memory statistics.
- Built a **4-level, system-call-aware MLFQ scheduler** with **round-robin queues**, adaptive time quanta, **CPU-bound process demotion**, and **periodic priority boosts** to prevent starvation.
- Implemented **demand-paging** support with **Clock-based page replacement, swap-in/swap-out handling**, per-process memory metrics, and **scheduler-aware victim selection**.
- Designed **disk-backed swap** with configurable **FCFS/SSTF request scheduling** and simulated **RAID 0, RAID 1, and RAID 5 mappings**, including **parity-based reconstruction** and disk-latency tracking.

**MIDI Piano Roll Viewer**

*C++ | SDL 2 | Bitwise Operations*

- Built a **C++ application that parses Standard MIDI binary files** and visualizes multi-track compositions as an interactive piano-roll timeline.
- Implemented **variable-length timestamp decoding, MIDI running-status handling**, and **note-on/note-off event matching** to reconstruct note durations.
- Developed zoom, pan, playhead animation, channel-based note coloring, and keyboard-grid rendering using SDL2.
- Designed **modular parser and renderer components**, supporting **MIDI formats 0 and 1** with robust malformed-file error handling.

**Machine Learning Library**

*Python | NumPy | Machine Learning | Neural Networks*

- Engineered a machine learning framework featuring a **custom autograd engine (Value class)** for **dynamic computational graph construction**, and integrated **Graphviz** to render forward and backward passes.
- Architected a **modular deep learning API** with **Sequential and Linear modules**, utilizing it to train the best-performing **Softmax Regression classifier** on the Fashion-MNIST dataset.
- Programmed a **Logistic Regression model** to perform spam classification, analyzing model convergence and error rates across raw and standardized data.
- Implemented custom data preprocessing (StandardScaler) and **K-Fold cross-validation** to automate **hyperparameter tuning** and optimize model generalization.

**Electric Motor Performance Optimization**

*Python | PyTorch | ANN | NSGA-II | Pymoo*

- Built an **ANN surrogate model (PyTorch)** to predict motor speed and torque from current/voltage inputs, achieving **$R^2 > 0.8$** across training, validation, and test sets.
- Formulated and solved a **multi-objective optimization problem** using NSGA-II (pymoo) to identify **Pareto-optimal operating points** balancing the speed-torque trade-off, using the trained ANN as the objective function.

- Generated and visualized the **Pareto front** across **100 generations (population size 500)**, demonstrating the trade-off structure between maximizing motor speed and torque for optimal operating configurations.

## SKILLS

- **Programming:** C, C++, Python, SQL
- **Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems
- **Libraries & Frameworks:** NumPy, SciPy, Matplotlib, Pandas, Scikit-learn, PyTorch
- **Tools:** Git, GitHub, Linux, Jupyter Notebook

## SCHOLASTIC ACHIEVEMENTS

- Ranked in the top 5.1% globally in a CodeChef contest, out of more than 18,800 participants.

## RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Data Structures & Applications, Operating Systems, Foundations of Machine Learning, Optimization Techniques I.
- **Math:** Calculus, Linear Algebra, Probability, Transform Techniques, Differential Equations, Complex Variables, Statistics.
- **Soft Skills:** Communication Skills, Introduction to Entrepreneurship.
- **Numerical Methods:** Fundamentals of Scientific Computing, Computer Aided Numerical Methods, Computational Methods in Material Science.